

Xiang MENG

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Education

PhD, Harvard University , Statistics	<i>Expected 2025</i>
Thesis committee: Natesh Pillai, Iavor Bojinov, Luke Miratrix	
MSc, University of Washington (UW) , Statistics	2020
BSc, National University of Singapore (NUS) , Quantitative Finance and Statistics	2018
Minor in Computer Science. Exchange program in University of California, San Diego (UCSD).	

Research Interest

My research develops causal inference methodologies designed for high-resolution data—information structures with granular detail that traditional statistical methods often simplify or ignore.

Selected Achievements & Awards

• Rising Stars in Data Science, selected participant recognized for innovative research and career potential, hosted by UC San Diego, University of Chicago, and Stanford University	2024
• Bok Teaching Certificate, Harvard University	2024
• Graduate School Conference Travel Award, UW	2019
• Lijen Industrial Development Medal for best thesis project in the discipline, NUS	2018
• Provost's Honors for academic excellence, Revelle College, UCSD	2017

Publications & Preprints (*indicates equal contribution)

[1] **Meng, X.**, Smith, A., Miratrix, L. (2025+). Variance estimation after matching or re-weighting. *In preparation. Preprint on the website.*

[2] **Meng, X.**, Bojinov, I. (2025+). Time-Varying Causal Survival Learning. *arXiv: 2503.00730.*

[3] Weingarden, H., **Meng, X.**, Armev, M., Onnela, J-P., Jaroszewski, A., Armstrong, C., & Wilhelm, S. (2025). Predicting the strength of next-day negative emotion states in body dysmorphic disorder using passive smartphone data: An intensive longitudinal assessment study. *Internet Interventions.*

[4] Che J. *, **Meng X. ***, Miratrix L. (2024+). Caliper Synthetic Matching. *arXiv:2411.05246.*

[5] **Meng X.**, Dempsey W., Liao P., Reid N., Klasnja P., Murphy S. (2024+). Evaluation of the HeartSteps Online Algorithm. *arXiv: 2501.02137.*

[6] Meng, Z., Yang, L., Raveendran, G., **Meng, X.**, Lin, J. (2024). Predict Progression Free Survival and Overall Survival Using Objective Response Rate for Anti-PD1/PDL1 Therapy Development. *BMC Cancer* **24**, 912 (2024). <https://doi.org/10.1186/s12885-024-12664-1>

[7] Wang L., **Meng X.**, Richardson T., Robins J. (2022). Coherent modeling of longitudinal causal effects on binary outcomes. *Biometrics*. 2022 May 4. doi: 10.1111/biom.13687.

[8] **Meng X.**, Huang, J. (2021). **REFINE2**: A tool to evaluate real-world performance of machine-learning based effect estimators for molecular and clinical studies. *arXiv: 2105.13148; Submitted to International Journal of Epidemiology.*

[9] Gordon E., **Meng X.**, Barnes M., Bhattacharjee T., & Srinivasa, S. (2019). Adaptive Robot-Assisted Feeding: An Online Learning Framework for Acquiring Previously Unseen Food Items. *International Conference on Intelligent Robots and Systems, Las Vegas, US, 2020*

[10] Gordon E., **Meng X.**, Barnes M., Bhattacharjee T., & Srinivasa, S. (2019). Learning from failures in robot-assisted feeding: Using online learning to develop manipulation strategies for bite acquisition. *IJCAI 2019 Workshop on AI × Food.*

[11] **Meng X.** (2018). Dynamic Mean-Variance Portfolio Selection. *Undergraduate Thesis. arXiv:1907.03093*

Software & Methodologies

Xiang Meng, Jonathan Che, Luke Miratrix. Caliper Synthetic Matching. R package. In Preparation

Linbo Wang, **Xiang Meng**, Thomas Richardson, James Robins. Methodology of “Coherent modeling of longitudinal causal effects on binary outcomes”. *Harvard Dataverse*, <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/UZAI96&faces-redirect=true>

Xiang Meng and Jonathan Huang. "REFINE2: A tool to evaluate real-world performance of machine-learning based effect estimators for molecular and clinical studies". *Github*, <https://github.com/mengeks/drml-plasmode>

Contributed Conference and Poster Presentations

Time-Varying Causal Survival Learning

American Causal Inference Conference (ACIC) May 2025

AAAI-25 AI for Causal Techniques Mar 2025

Harvard Dept. Government Applied Statistics Workshop Nov 2024

Rising Star Workshop, UCSD Nov 2024

Caliper Synthetic Matching:

Peking University School of Education invited talk Jan 2025

New England Statistics Symposium (NESS) May 2024

American Causal Inference Conference (ACIC) May 2024

New England Student Research Symposium on Statistics and Data Science Apr 2024

A Novel Evaluation of a Just-in-Time Adaptive Intervention Algorithm:

INFORMS Annual Meeting Oct 2023

Harvard Statistics Department Seminar Oct 2022

Eastern North American Region (ENAR) Meeting Mar 2022

14th CMStatistics, Advances in statistical methods for mobile health Dec 2021

3rd Harvard Health Data Science Symposium Nov 2021

Joint Statistical Meeting (JSM) Aug 2021

Causal Questions in Micro-Randomized Trials (MRTs): Introduction and Challenges:

Society for Causal Inference (SCI) Causal Inference for Social Impact Jun 2021

Learning from failures in robot-assisted feeding:

IJCAI 2019 Workshop on AI×Food, Poster and Oral Presentation Aug 2019

Teaching Experience

Introduction to Bioinformatics, Harvard University, Head Teaching Fellow (TF). Designed weekly section materials. Spring 2025

Introduction to Data Science, Harvard University, Head Teaching Fellow (TF) for 220 students, oversaw 29 TFs. Fall 2023

The Art of Teaching in Statistics, Harvard University, teaching preparation for all 1st year statistics PhD students. AY 2022 – 23

Data Analysis in Modern Biostatistics (Biomarker and Cancer Research), Harvard University Spring 2022

Introduction to Probability, Harvard University Fall 2021

STEMPREP Summer Statistics Course for 7th and 8th Grade Students, UW Summer 2019

Programming Methodology in Python, NUS AY 2015 - 16, Spring 2018

Academic & Student Service

Reviewer for *Biometrika*, *Biometrical Journal*, *Journal of Applied Statistics (JAS)*, *Americal Causal Inference Conference (ACIC)*, *NeuroIPS* workshop.

Co-Chair for seminar department at Harvard Chinese Student & Scholar Association (HCSSA), organized and supervised 7 seminars to give industry and academia exposure to Chinese graduate students across art and science, industry and academia.

Volunteered experiences (10+ times) in supporting elders, mentally disables, women and gender-diverse individuals.

Work Experience

Harvard University

Sep 2021 - Present

Graduate Research Assistant

- Develop causal inference methods integrating machine learning for large-scale data in public health, education, and business to improve decision-making and personalize interventions.
- Publish peer-reviewed research and present at conferences in collaboration with interdisciplinary teams.

Sanofi

May – Aug 2022

Summer Intern, Clinical Evidence and Data Science

- Co-created a comprehensive database from over 400 clinical trials on Anti-PD1/PDL1 cancer treatments using R.
- Co-developed a predictive method for three survival outcomes in cancer patients.

Dymon Asia Capital

April – July 2017

Summer Intern, Risk Analysis

- Improved risk reporting procedure for macro fund using VBA; shortened procedure from 2 hours to less than 20 seconds.
- Streamlined daily risk management by meticulously monitoring daily equity fund risk and macro fund risk.

Skills

Languages: Mandarin Chinese (Native), English and Japanese (Proficient in writing, speaking, reading, listening)

Programming & IT: Python, SQL, R, SAS, SPSS, C++, Unix, Java, VBA, Matlab